SYSDESIGN REFERENCE

Database Sharding Tradeoff Matrix

A decision-oriented comparison of alternatives and failure modes.

Partition data so storage and throughput scale beyond one database boundary.

VERSION 1.0 REVIEW
REVIEWED THROUGH 2026-07-18
STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / DATABASE-SHARDING

SYSDESIGN

Database Sharding Tradeoff Matrix

CORE IDEA

Partition data so storage and throughput scale beyond one database boundary.

Reference architecture decision chain

01	02	03	04
Shard Key	Routing	Rebalancing	Cross-Shard Queries

Decisions to defend

+ Shard key
+ routing
+ rebalancing
+ cross-shard queries
+ and tenant isolation.

Failure modes to test

+ Hot shards
+ skew
+ resharding
+ and distributed transactions.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Database Sharding Tradeoff Matrix

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

PUBLISHING STATUS

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